OFFICIAL OPERATIONS MANUAL • ENTERPRISE EDITION

NETSPECTRE

Visual Network Intelligence, Multi-Protocol Diagnostics & Endpoint Troubleshooting Platform

---

Comprehensive user manual and technical guide for NETSPECTRE V3.1 Enterprise. Covering the Tri-Persona Workspace system (Personal, Developer, Network Operations), native Win32 kernel telemetry, dual-mode packet capture, Chromium browser observability extension, persistent data quotas, and deep troubleshooting suites.

APPLICATION VERSION	TARGET PLATFORM	CAPTURE ENGINES	SECURITY MODEL
v3.1 ENTERPRISE	Windows 10 / 11 x64	Win32 IP Helper & Mode 2 Raw	Localhost Token IPC

1. EXECUTIVE OVERVIEW & PERSONA WORKSPACES

NETSPECTRE V3.1 is an enterprise-grade Windows desktop application that bridges the gap between low-level system packet dynamics and user-facing actionable intelligence. Operating on a **100% genuine Win32 telemetry foundation**, NETSPECTRE queries the Windows IP Helper API (`iphlpapi.dll`), kernel process handles, and network adapter hardware counters without simulation.

The Tri-Persona Workspace Architecture

NETSPECTRE V3.1 eliminates interface clutter by tailoring tools and views to three distinct user personas via the top-bar workspace switcher:

👤 Personal Workspace

Everyday Windows Users & Remote Workers: 1-Click Network Health Doctor (0-100 Quality Index), Explain My Network Assistant ("Why is my internet slow?"), persistent daily/monthly bandwidth quotas, background process trackers, and 1-click Windows Defender Firewall application blocking with emergency rollback.

Developer Workspace

Software & API Engineers: Interactive REST API Debugger with microsecond phase waterfalls, Live Browser Observability via Chrome/Edge extension, HAR archive importer, Failure Root-Cause Analyzer, and multi-stage Environment Health Monitoring (DEV/SIT/UAT/PROD).

🌐 Network Operations Workspace

IT Administrators, Network Engineers & SOC Analysts: Dual-Mode Packet Capture (Mode 1 Basic Telemetry vs Mode 2 Deep Wire Sniffer with genuine PCAP export), 2.5D Orthographic Neural Highway, deep Troubleshooting Workbench (Ping, Traceroute, 7-record DNS, Path MTU, Port Prober, TLS Inspector, Routing Table), Incident Forensics Recorder with HTML export, Smart Alerts Engine with Windows Desktop Notifications, and Multi-Machine Monitor.

Core Operating Principles

- **100% Genuine Win32 Telemetry:** Every displayed socket, PID, protocol, and throughput metric reflects actual kernel data. No synthetic frames or simulated connections.
- **Dual-Mode Honest Capture (REQ-01):** Mode 1 provides connection flows. Mode 2 provides wire packet frames when running elevated as Administrator. PCAP export exports strictly captured packets.
- **Local-First Enclave Security (REQ-15):** All telemetry stays on your machine. The internal REST engine enforces cryptographically secure session token verification (`X-NetSpectre-Auth`) and CORS validation.

2. PERSONAL WORKSPACE — DOCTOR & ASSISTANT

The **Network Doctor** (`network-doctor`) provides an automated health checkup of your workstation's complete network path, evaluating local router reachability, public internet backbone latency, DNS resolution speed, and physical Wi-Fi signal quality.

Health Score & Quality Index (0–100)

SCORE RANGE	STATUS RATING	COLOR INDICATOR	OPERATIONAL MEANING
85 – 100	EXCELLENT	Emerald Green	Optimal link quality. Latency < 50ms, DNS < 60ms, Wi-Fi > 75%. Ready for competitive gaming & 4K streaming.
65 – 84	GOOD	Cyan Sky	Normal web browsing and video conferencing. Moderate latency without noticeable dropped packets.
40 – 64	DEGRADED	Amber Orange	High latency or weak Wi-Fi signal. Video buffering and sluggish API calls expected.
0 – 39	CRITICAL	Rose Red	Gateway unreachable, internet offline, or DNS lookup failure. Network intervention required.

Explain My Network Assistant (REQ-08)

Integrated into the Network Doctor is the **Explain My Network** deterministic intelligence assistant. Users can ask questions in plain English or select from quick diagnostic prompts:

- "*Why is my internet slow?*" — Pinpoints gateway congestion, high ISP backbone latency, or heavy background consumers.
- "*Which applications consume my bandwidth?*" — Lists top processes moving data with exact byte counts.
- "*Where is my data going?*" — Identifies remote countries and ASNs receiving outbound traffic.
- "*Is DNS resolving correctly?*" — Diagnoses NXDOMAIN or slow DNS response times.
- "*Is a firewall blocking my connections?*" — Checks active block rules and connection rejection rates.

Every diagnosis delivers verified **Telemetry Evidence**, an objective confidence rating, and concrete **Remediation Steps**.

3. PERSISTENT USAGE QUOTAS & FIREWALL CONTROLS

Persistent Internet Usage & Quota Budgeting (REQ-03, REQ-16)

The **Internet Usage Monitor** (`internet-usage`) tracks per-process bandwidth consumption with persistent local transactional storage in `%APPDATA%\NetSpectre\data\usage.json`:

- **Daily & Monthly Quotas:** Configurable data caps (e.g., 5 GB/day, 50 GB/month) with visual progress bars and warning alerts.
- **Persistent Daily Session Archive:** Retains cumulative bandwidth metrics per application across system reboots.
- **Stealth Background Consumer Detection:** Identifies silent background processes moving high-volume data without open windows.
- **Telemetry & Tracker Audit:** Detects connections communicating with ad, analytics, and telemetry endpoints.

Application Internet Firewall Controls (REQ-04)

The **Firewall Manager** (`firewall`) allows users to restrict outbound network access for any Windows executable with a single click using native Windows Defender Firewall rules:

```
netsh advfirewall firewall add rule name="NETSPECTRE Block - [App]" dir=out action=block program="[Path]"
enable=yes
```

Administrator Elevation Check

Automatically checks elevation via `WindowsBuiltInRole.Administrator`. If non-elevated, displays a clear read-only banner preventing silent command failures.

Emergency Rollback All Rules

1-Click safe rollback instantly removes all NETSPECTRE-managed firewall block rules, immediately restoring outbound internet connectivity if needed.

4. API INSPECTOR & BROWSER OBSERVABILITY

Live Browser Observability Extension (REQ-02)

NETSPECTRE includes a built-in Chromium extension in `extension/` for Google Chrome, Microsoft Edge, and Brave:

- **Installation:** Open `chrome://extensions`, enable Developer Mode, and click "*Load unpacked*" pointing to the `extension/` folder.
- **Automatic Secret Redaction:** Automatically masks `Authorization`, `Bearer`, `Cookie`, and API key tokens before telemetry leaves the browser.
- **Live Feed:** Forwards request URLs, HTTP methods, status codes, duration, and payload sizes directly into the NETSPECTRE API Inspector.

Direct API Request Executor & Phase Waterfall

Execute GET, POST, PUT, DELETE, PATCH, and HEAD requests with headers and JSON payload editors. Every request is broken down into microsecond phases:

```
Total Duration = DNS Lookup + TCP Connect + TLS Handshake + TTFB + Content Download
```

HAR Archive Importer

Import any `.har` (HTTP Archive) export from Chrome DevTools, Postman, or Charles Proxy to inspect external request cascades, response codes, and timings offline.

5. MULTI-TIER ENVIRONMENT HEALTH MONITOR

The **Environment Health Monitor** (`env-monitor`) provides continuous reachability and SSL certificate health telemetry across software deployment pipelines:

DEV (Development)

Local microservices and containerized backends (e.g., `127.0.0.1:48123` or local Docker containers).

SIT (System Integration)

Internal integration testing endpoints validating API contract compatibility.

UAT (User Acceptance / Staging)

Pre-production environments matching production configuration.

PROD (Production)

Live customer-facing microservices and edge gateways.

Persistent Endpoint CRUD (REQ-05)

- **Add Custom Endpoints:** Specify display name, target host/URL, and environment tier.
- **Persistent Storage:** Endpoints are persisted in `%APPDATA%\NetSpectre\data\environments.json` across restarts.
- **SSL Expiry Countdown:** Continuously monitors TLS certificate remaining validity days to prevent unexpected production outages.
- **1-Click Workbench Integration:** Launch TLS Certificate Inspection or Ping probes directly from any endpoint card.

6. DUAL-MODE PACKET CAPTURE & PCAP

Mode 1 vs Mode 2 Capture Semantics (REQ-01)

NETSPECTRE provides strict cryptographic truthfulness regarding packet capture capabilities:

FEATURE	MODE 1: BASIC TELEMETRY	MODE 2: ADVANCED PACKET CAPTURE
Elevation	Standard User (Non-Elevated)	Administrator Elevation Required
Capture Engine	Win32 IP Helper (iphlpapi.dll)	Raw Socket Wire Sniffer (SIO_RCVALL / PktMon)
Process Attribution	Exact Kernel PID & Image Path	Exact Socket Attribution
Raw Wire Headers	Not available (socket metadata only)	Full Ethernet/IP/TCP/UDP Wire Frames
PCAP Export	Disallowed (Prevents fake frames)	Genuine Binary PCAP for Wireshark

Genuine PCAP Export

When Mode 2 capture is active, clicking **"Export Captured PCAP"** produces a standard binary PCAP file (`netspectre-capture.pcap`) containing exclusively captured wire frames for Wireshark inspection.

7. TROUBLESHOOTING WORKBENCH (DNS & MTU)

Expanded DNS Inspector (7 Record Types) (REQ-06)

The DNS Inspector supports querying all standard DNS record types with lookup timing:

RECORD TYPE	PURPOSE	EXAMPLE OUTPUT
A	IPv4 Host Address	142.250.190.46
AAAA	IPv6 Host Address	2607:f8b0:4004:800::200e
CNAME	Canonical Alias Name	github.map.fastly.net
MX	Mail Exchange Server	aspmx.l.google.com (Priority 10)
TXT	SPF, DKIM & Domain Verification	v=spf1 include:_spf.google.com ~all
NS	Authoritative Nameserver	ns1.cloudflare.com
SOA	Start of Authority & Zone Serial	ns-cloud-a1.googledomains.com

Path MTU Discovery (REQ-06)

The **Path MTU Discovery** tool probes upstream packet size limits using Don't Fragment (DF=1) ICMP echo probes:

- Identifies MTU bottlenecks caused by VPN tunnels, PPPoE encapsulation, or cellular uplinks.
- Probes across sizes 1200 B to 1500 B to discover the exact maximum transmission unit without fragmentation.

8. INCIDENT FORENSICS & SMART ALERTS

Incident Forensics Recorder (REQ-07, REQ-11)

Freeze real-time network state during an outage or latency burst to generate complete post-mortem evidence:

- **Sliding Telemetry Buffer:** Captures socket states, process consumption, and link health at the moment of trigger.
- **Persistent Incident Archive:** Saves all recorded incidents in `%APPDATA%\NetSpectre\data\incidents.json` for historical comparison.
- **Multi-Format Exports:** Export incident reports as **Standalone HTML**, **Markdown**, or **JSON**.

Smart Alerts Engine & Windows Notifications (REQ-09)

- **Configurable Thresholds:** Set custom latency spike limits (50–500 ms) and outbound burst limits (1–50 MB/s).
- **Windows Desktop Notifications:** Dispatches native notifications via the Electron bridge when anomalies are detected.
- **Persistent Alert History:** Stored in `%APPDATA%\NetSpectre\data\alerts.json` with individual acknowledgement buttons.

Multi-Machine Endpoint Monitor (REQ-12)

Centralized monitoring for local default gateways, DNS clusters, and enrolled remote Windows endpoints:

```
NetSpectreEngine.exe --port 48123 --token <SHARED_SECRET>
```

9. REST API REFERENCE & OPERATIONS FAQ

METHOD	ENDPOINT PATH	DESCRIPTION
GET	/api/status	Engine health and version disclosure.
GET	/api/telemetry	Full kernel telemetry snapshot (requires X-NetSpectre-Auth).
GET	/api/capture/status	Mode 1 vs Mode 2 status and packet counters.
POST	/api/capture/start	Starts Mode 2 raw socket wire capture (Administrator only).
GET	/api/pcap/export	Exports binary PCAP stream of captured wire frames.
POST	/api/browser-obs/ingest	Ingests HTTP request telemetry from Chromium extension.
GET	/api/usage/history	Persistent daily and monthly bandwidth usage and quotas.
POST	/api/usage/limits	Sets daily and monthly quota caps in MB.
POST	/api/diagnose/explain	Deterministic natural language network diagnostic engine.
GET	/api/workbench/mtu	Discovers Path MTU using Don't Fragment ICMP probes.

Operations FAQ

Q: Why does Mode 2 Packet Capture require Administrator elevation?
A: Windows security architecture restricts raw socket binding (`SIO_RCVALL`) strictly to processes running with Administrator privileges to protect system integrity.

Q: Are browser requests encrypted when captured by the extension?
A: The Chromium extension observes request metadata before transport encryption, allowing inspection of URLs, HTTP methods, headers, and status codes while masking credentials.

Q: Where are my usage history and incident recordings stored?
A: All persistent data is stored locally in `%APPDATA%\NetSpectre\data\` in transactional JSON files. No data is ever transmitted to external servers.